CHAPTER 3: RESULTS

3.1 Demographic and Occupational Profile

This chapter presents the study findings in a structured sequence, beginning with descriptive results that characterise the demographic and occupational profile of the sample and the prevalence of each health outcome. Analytic results — including bivariate associations and multivariate regression models — are presented subsequently to contextualise the descriptive findings within a risk-factor framework.

A total of 376 wood industry workers participated in the study, representing an 87.5% response rate from 430 initially approached. The mean age of participants was 34.69 ± 11.46 years (range: 18–60 years), and the mean duration of employment in the wood industry was 11.74 ± 9.67 years, indicating a sample that ranged from relatively recent entrants to seasoned professionals with substantial cumulative occupational exposure.

Table 3.1.1: Socio-demographic profile of Study Participants

Variable	Category	n	Percentage %
Smoking status	Yes	91	24.2
	No	285	75.8
Education	Illiterate	95	25.3
	Primary	113	30.1
	Secondary	144	38.8
	Higher	24	5.1
Marital Status	Single	104	27.7
	Married	272	72.3
PPE usage	Never	124	33.0
	Sometimes	161	42.8
	Always	91	24.2
Workplace ventilation	Present	315	83.8
	Absent	59	15.7

The majority of participants had attained modest levels of formal education: 38.8% had completed secondary schooling, 30.1% primary education, 25.3% were entirely illiterate, and a mere 5.1% had achieved higher education. This educational profile carries direct occupational safety implications, as limited literacy may impair workers' capacity to interpret hazard warning labels, follow written safety protocols, and engage effectively with formal training programs. Approximately one in four workers (24.2%) were current smokers — a proportion of particular significance given the synergistic respiratory risks conferred by simultaneous tobacco and wood dust exposure.

Regarding institutional safety infrastructure, 58.8% of workers reported having received formal occupational safety training, and 58.0% had been provided with PPE. However, actual PPE utilization was markedly inconsistent: only 24.2% reported always using protective equipment, 42.8% used it sometimes, and 33.0% never used it. This pronounced compliance gap between provision and consistent utilization represents a critical programmatic finding. It highlights that physical availability alone cannot overcome behavioral or ergonomic barriers to safety compliance, such as equipment discomfort or a workplace culture that normalizes risk-taking . The majority of workplaces (83.8%) were reported to have some form of ventilation system, while 15.7% of workers operated in poorly ventilated environments. For these individuals, the absence of primary engineering controls maximizes continuous breathing-zone exposure to suspended particulates, directly elevating their localized occupational risk.

Table 3.1.2: Occupational Characteristics of Study Participants

Job Role	n	Percentage %
Carpenter	139	37.0%
Spray Painter	24	6.4%
Machine Operator	90	23.9%
Foreman	26	6.90%
Loader	30	8.0%
Polisher	67	17.8%

Table 3.1.2 delineates the distribution of primary job roles among the 376 wood industry workers sampled in this study. The workforce was predominantly engaged in direct, high-exposure manufacturing tasks. Carpenters constituted the largest occupational subgroup, representing over one-third of the sample (37.0%, n=139), followed by machine operators (23.9%, n=90) and wood polishers (17.8%, n=67). Together, these three primary roles accounted for nearly 80% of the total workforce, highlighting the heavily manual, mechanized, and chemically intensive nature of the woodworking operations at the Hayatabad Industrial Estate. General support and material handling staff, categorized as loaders, represented 8.0% (n=30) of the workers. Specialized and supervisory roles made up the remainder of the cohort, including foremen (6.9%, n=26) and spray painters (6.4%, n=24). This distinct occupational distribution provides critical baseline context for the diverse hazard profiles—ranging from heavy particulate dust and high-decibel noise to chemical vapors and ergonomic strain—that uniquely characterize this industrial setting.

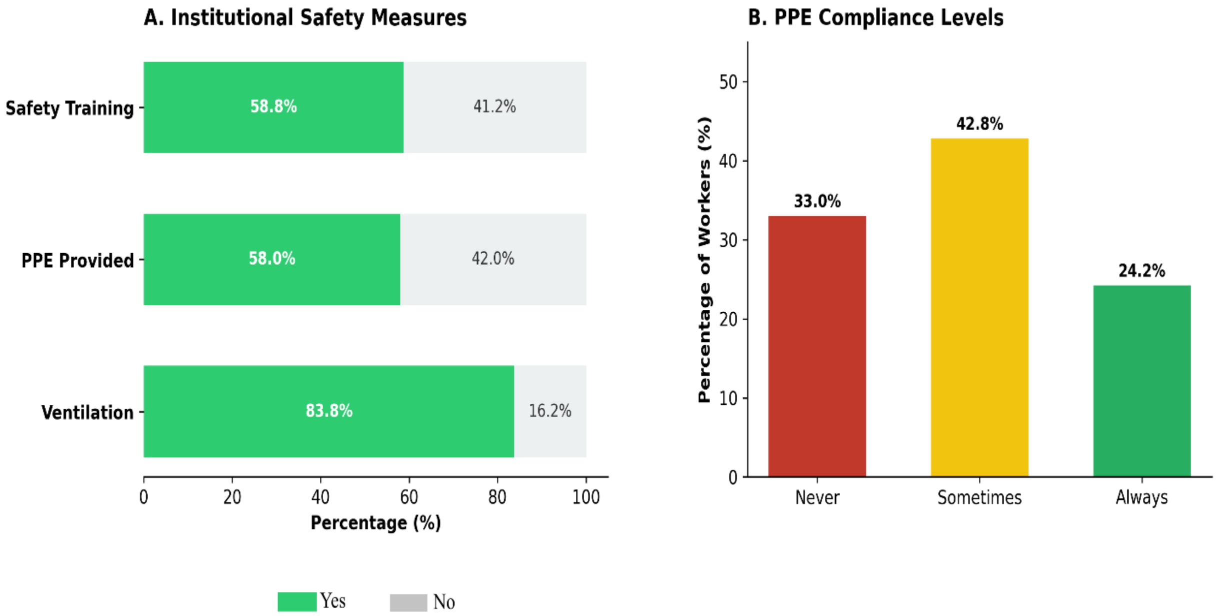

Figure 4: Institutional Safety and PPE Compliance

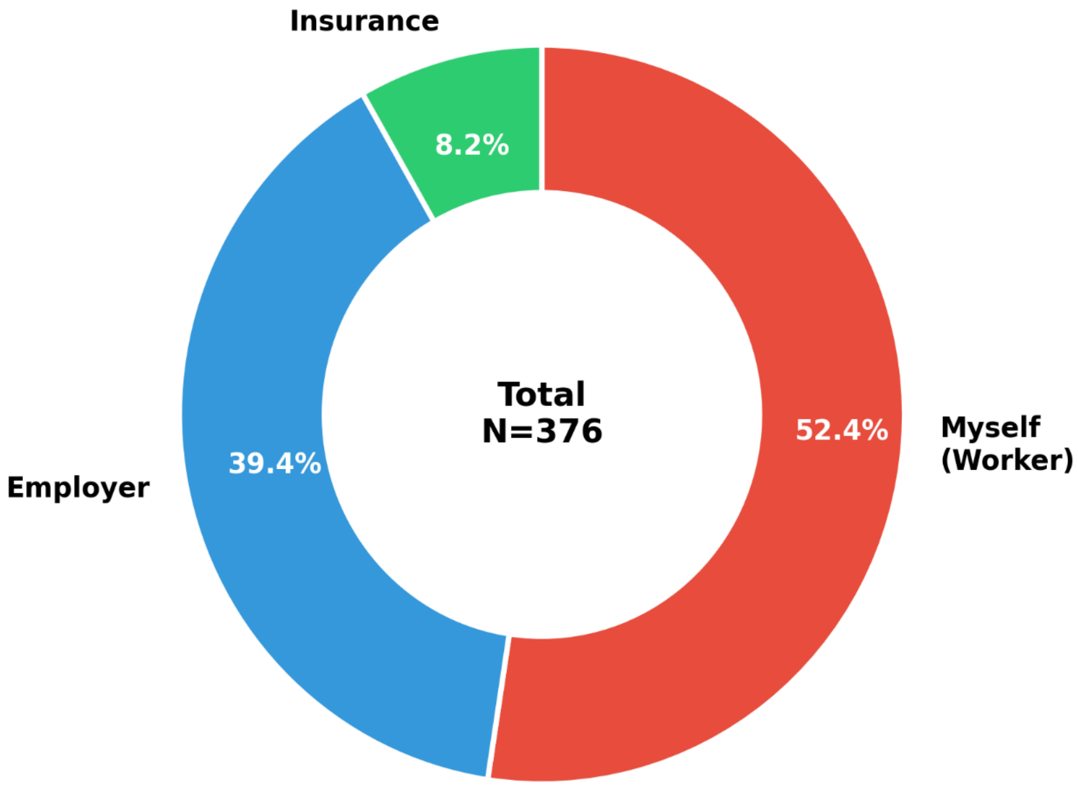

Figure 5: Medical Treatment Coverage

The demographic profile of this cohort highlights a profound socioeconomic vulnerability that compounds their occupational health risks. Notably, the financial precarity of this demographic is starkly reflected in their healthcare financing; a majority of the workforce (52.4%) bears the cost of medical treatment entirely out-of-pocket, with only a negligible fraction (8.2%) possessing formal health insurance. In an environment where over half the workforce has required time off due to occupational trauma, this lack of financial protection creates a vicious cycle where workplace morbidity directly precipitates economic hardship. Laborer continue to operate heavy machinery and endure particulate exposure despite actively suffering from chronic musculoskeletal pain or respiratory distress, simply out of financial necessity. Ultimately, this demographic snapshot reveals a workforce that is not only highly exposed to industrial hazards, but also systematically underequipped both educationally and economically to advocate for safer conditions or access restorative healthcare.

3.2 Hazard Perception, Safety Culture, and Injury Prevalence

Table 3.2: Hazard Perception, Safety Culture, and Injury Prevalence

Variable	Category	Frequency (n)	Percentage (%)
Hazard Awareness: Dust	Yes	288	76.6
	No / Unsure	88	23.4
Hazard Awareness: Noise	Yes	233	62
	No / Unsure	93	24.7
Machine Safety Guards	Yes, machinery has adequate guards	235	62.5
	No, machinery lacks safety guards	115	30.6

	Not applicable to my job	26	6.9
Ergonomic Autonomy (Breaks)	Yes, can take breaks without problem	141	37.5
	Yes, but must inform supervisor	175	46.5
	No, breaks are only at fixed times	50	13.3
	No, there are very few breaks	10	2.7
Confidence Reporting Safety	Very confident	116	30.9
	Somewhat confident	170	45.2
	Not at all confident	90	23.9
Time Off Due to Work Injury	Yes	216	57.4

To better understand the behavioral and organizational context of the Hayatabad Industrial Estate, workers were assessed on their perception of occupational hazards, workplace safety culture, and recent injury history (Table 3.2).

A considerable proportion of the workforce demonstrated a high awareness of specific physical hazards. The majority of workers (76.6%) correctly identified wood dust as a significant health risk. However, awareness regarding auditory hazards was notably lower, with only 62.0% recognizing the harmful effects of occupational noise. This leaves over a third of the workforce either unaware or unsure of noise-induced risks.

Despite general hazard awareness, the institutional safety culture revealed significant vulnerabilities. When asked about their confidence in reporting safety problems or hazards to management, only 30.9% of workers felt "Very confident." The largest segment (45.2%) felt only "Somewhat confident," and nearly one-quarter of the workforce (23.9%) reported feeling "Not at all confident" in bringing safety issues to light.

This precarious safety culture is reflected in a high rate of occupational trauma; more than half of the surveyed workforce (57.4%) reported having to take time off work due to a workplace injury.

Assessment of the physical and ergonomic workplace environment revealed critical gaps in primary safety infrastructure. Nearly one-third of the workforce (30.6%) reported operating machinery that lacked adequate safety guards or protective features. Regarding ergonomic autonomy and physical recovery, only 37.5% of workers felt they could take short breaks freely without issue. The largest proportion of the workforce (46.5%) was required to obtain explicit supervisory permission to rest, and 16.0% operated under rigid or highly restricted break schedules. This lack of ergonomic autonomy substantially compounds the cumulative musculoskeletal loading observed in this cohort

3.3 Quality of Life Impact

Table 3.3: Impact of work-related health problems on the personal and home life of wood industry workers.

Impact on Quality of Life	Percentage of Workers
Completely Unaffected	31.90%
Slight Reduction	61.20%
Significantly Impaired	6.90%

The occupational morbidity documented in this study translated directly into functional limitations outside the workplace. When asked if their work-related health problems affected their ability to perform household chores or enjoy their personal life, a striking majority reported a negative impact. Specifically, 61.2% of workers experienced a 'slight' reduction in their quality of life, while 6.9% reported that their occupational injuries 'significantly' impaired their personal and home life. Only 31.9% of the workforce reported that their home life remained completely unaffected by their occupational health status

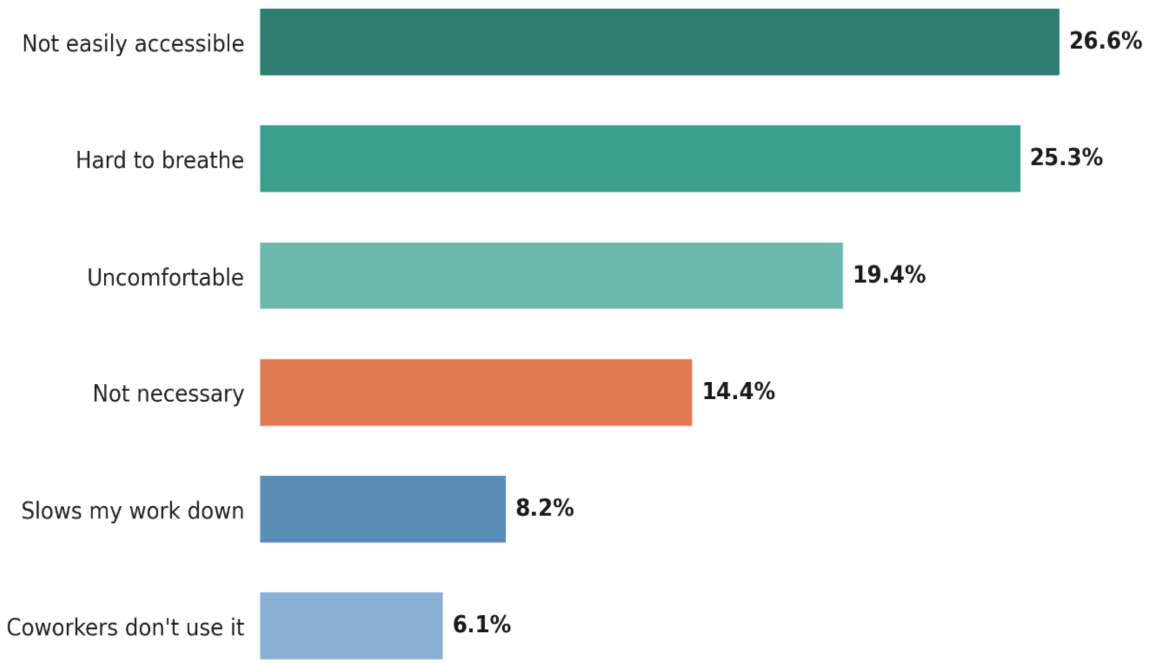

Figure 6: Primary Barriers to PPE Utilization Among Woodworkers

The self-reported barriers to PPE utilization demonstrate that non-compliance among woodworkers stems primarily from structural and ergonomic failures rather than a lack of hazard awareness. The leading obstacles—equipment being physically inaccessible at the workstation (26.6%) and causing acute physiological distress, such as breathing difficulties (25.3%) and discomfort (19.4%) —reveal that workers are often forced to prioritize immediate operational viability over protective measures. Because behavioral factors like risk normalization (14.4%) remain secondary, occupational health interventions must shift away from basic health education and instead focus on providing ergonomically appropriate, breathable safety gear directly at the point of use

3.4 Prevalence of Occupational Health Problems

Health problem were dichotomized into Low Problem Burden (Never or Rarely reported) and High Problem Burden (Sometimes or Often reported) to facilitate clinical interpretation and statistical analysis. The distribution of problem frequencies across all six health domains measured in this study is presented in Figure 6.

Musculoskeletal pain emerged as the most prevalent occupational health problem, with 38.0% of workers (n = 143) reporting high-frequency problems. Respiratory cough was the second most prevalent condition (34.8%, n = 131), followed by headaches (27.1%, n = 102), eye irritation (23.9%, n = 90), skin problems (21.0%, n = 79), and tinnitus (17.6%, n = 66). The simultaneous co-occurrence of multiple problems within individual workers is consistent with the multi-hazard and cumulative exposure environment characteristic of wood processing settings, and underscores the importance of adopting comprehensive — rather than single-hazard — occupational health surveillance strategies.

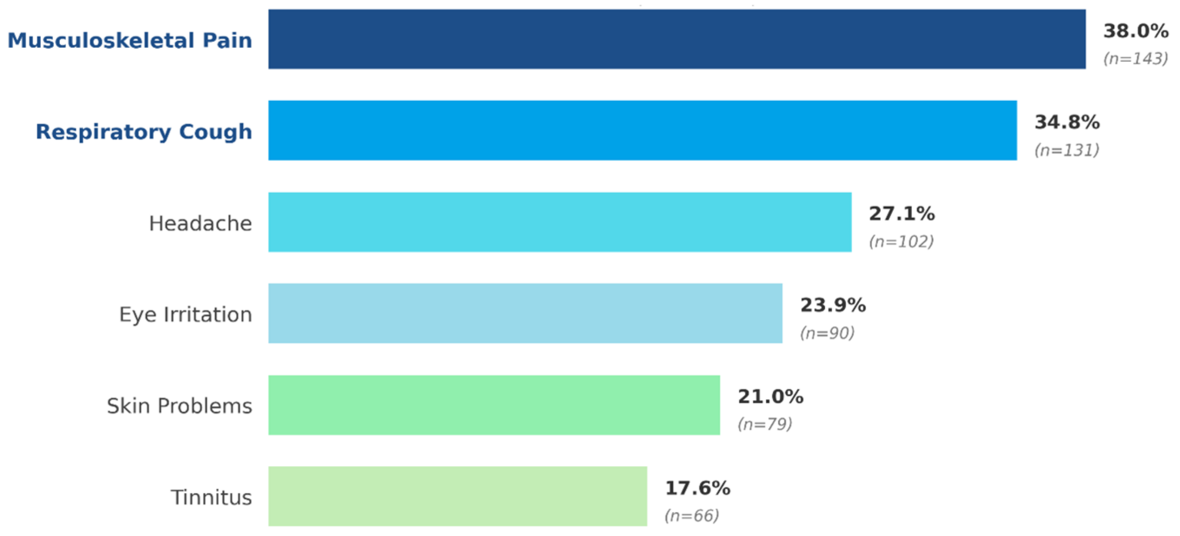

Figure 7: Prevalence of occupational health Problems

3.5 Association Between PPE Usage and Health Problems

Chi-square analyses were performed to assess the statistical association between PPE usage frequency and each of the six health outcomes.

Table 3.5: Chi-Square Analyses — Association Between PPE Usage Frequency and Health Outcomes

Health Outcome	PPE: Never	PPE: Sometimes	PPE: Always	Total	Problem Frequency	χ^2	df	p
Respiratory Cough	57	128	60	245	Low	34.733	2	**< 0.001**
	67	33	31	131	High			
Eye Irritation	84	126	76	286	Low	7.920	2	**0.019**
	40	35	15	90	High			
Skin Problems	85	134	78	297	Low	12.369	2	**0.002**
	39	27	13	79	High			
Musculo-skeletal Pain	66	102	65	233	Low	7.608	2	**0.022**
	58	59	26	143	High			
Tinnitus	88	139	83	310	Low	17.800	2	**< 0.001**
	36	22	8	66	High			
Headache	87	122	65	274	Low	1.244	2	**0.537**
	37	39	26	102	High			

PPE usage demonstrated statistically significant associations with five of the six health outcomes examined in this study, with the sixth — headache — falling short of statistical significance ($\chi^2 = 1.244$, $p = 0.537$). Among the significant associations, the strongest effect was observed for respiratory cough ($\chi^2 = 34.733$, $p < 0.001$), followed by tinnitus ($\chi^2 = 17.800$, $p < 0.001$), skin problems ($\chi^2 = 12.369$, $p = 0.002$), eye irritation ($\chi^2 = 7.920$, $p = 0.019$), and musculoskeletal pain ($\chi^2 = 7.608$, $p = 0.022$). Across all five significant outcomes, the distribution of Problem severity shifted notably toward lower frequency among workers reporting consistent PPE use, which lends face validity to the protective role of personal protective equipment in this occupational setting.

It is important, however, to interpret these bivariate associations with appropriate methodological caution. Chi-square analyses, by their nature, establish co-occurrence rather than causation, and the cross-sectional design of this study precludes any inference about temporality or directionality. A particular concern here is the possibility of reverse causality: workers assigned to higher-risk job roles — and thereby experiencing a greater problem burden — may simultaneously be issued and observed wearing PPE more frequently as a direct consequence of institutional risk stratification or supervisory oversight. In other words, the relationship between PPE use and problem severity may, in part, run in the opposite direction to what a straightforward protective interpretation would suggest. Additionally, unmeasured confounders such as duration of occupational exposure, workshop ventilation quality, and pre-existing health conditions could plausibly explain some portion of the observed variation. These limitations inherent to bivariate analysis underscore the necessity of the multivariate regression models presented in Section 3.6, which allow for simultaneous adjustment across relevant covariates and provide more precise estimates of association, though it is acknowledged that causal inference remains precluded by the cross-sectional design.

3.6 Association Between Job Role and PPE Usage

To examine these categorical variables, a chi-square analysis was performed to assess the statistical association between the workers' specific job roles and their self-reported frequency of personal protective equipment (PPE) usage

Table 3.6: Chi-Square Analysis — Association Between Job Role and PPE Usage.

Symptom	Burden	Carpenter	Painter	Machine Operator	Foreman	Loader	Polisher	χ²	P-Value
Cough	High	45	6	35	11	8	26	9.7	0.082
	Low	94	18	55	15	22	41		
Eye	High	26	8	20	7	8	21	2.4	0.787
	Low	113	16	70	19	22	46		
Skin	High	24	6	20	7	9	13	4.3	0.507
	Low	115	18	70	19	21	54		
MSK	High	53	12	21	14	12	31	8.7	0.12
	Low	86	12	69	12	18	36		
Tinnitus	High	19	5	17	5	6	14	2.1	0.823
	Low	120	19	73	21	24	53		
Headache	High	36	9	19	8	5	25	4.6	0.458
	Low	103	15	71	18	25	42		

Chi-square analyses revealed no statistically significant association between workers' specific job roles and any of the six occupational health outcomes examined. Cough showed the strongest trend across job categories ($\chi^2 = 9.757$, $p = .082$), followed by musculoskeletal pain ($\chi^2 = 8.733$, $p = .120$), though neither reached conventional significance. Eye irritation ($\chi^2 = 2.427$, $p = .787$), tinnitus ($\chi^2 = 2.184$, $p = .823$), headache ($\chi^2 = 4.667$, $p = .458$), and skin problems ($\chi^2 = 4.303$, $p = .507$) all demonstrated non-significant distributions across carpenter, painter, machine operator, supervisor, loader, and polisher roles. These findings suggest that occupational health burden reflects shared exposure conditions — such as workshop ventilation status, dust levels, and PPE behaviour — that cut across role boundaries.

3.7 Association Between Smoking and Health Problems

Table 3.7: Chi-Square Analysis — Association Between Smoke and Health Problems

Outcome	Ventilation	Low	High	X2	P-Value
Cough	Yes	49	42	6.769	0.009**
	No	196	89		
Eye Irritation	Yes	72	19	0.616	0.432 ns
	No	214	71		
MSK	Yes	54	37	0.352	0.553 ns
	No	179	106		
Tinnitus	Yes	79	12	1.582	0.209 ns
	No	231	54		
Headache	Yes	65	26	0.127	0.722 ns
	No	209	76		
Skin Problems	Yes	72	19	0.001	0.971 ns
	No	226	59		

Chi-square analyses examining the association between smoking status and the six occupational health outcomes revealed a statistically significant association only for respiratory cough ($\chi^2 = 6.769$, $p = 0.009$). Smokers were disproportionately represented in the high cough burden group compared to non-smokers, consistent with the well-established role of tobacco smoke as a respiratory irritant that compounds the effects of occupational dust and chemical exposures. All other outcomes — eye irritation ($p = 0.432$), musculoskeletal pain ($p = 0.553$), tinnitus ($p = 0.209$), headache ($p = 0.722$), and skin problems ($p = 0.971$) — showed no statistically significant variation by smoking status. These non-significant findings suggest that for most health outcomes in this cohort, occupational exposures such as dust, noise, and inadequate PPE usage may be more proximal determinants of morbidity than personal smoking behaviour, though residual confounding cannot be excluded given the cross-sectional design.

3.8 Association Between Ventilation Status and Health Problems

Table 3.8: Chi-Square Analysis — Association Between Workplace Ventilation and Health Problems

Outcome	Ventilation	Low	High	χ²	(p-value)
Cough	Present	218	97	16.83	(0.001)**
	Absent	25	34		
Eye Irritation	Present	239	76	0.870	(0.647) ns
	Absent	46	59		
MSK	Present	197	118	0.341	(0.843) ns
	Absent	35	24		
Tinnitus	Present	257	51	1.237	(0.539) ns
	Absent	58	8		
Skin Problems	Present	258	63	1.415	(0.493) ns
	Absent	44	15		
Headache	Present	231	84	0.648	(0.723) ns
	Absent	42	17		

Workers in poorly ventilated workshops reported significantly higher frequencies of respiratory cough problems ($\chi^2 = 22.92$, $p = 0.001$), while the association between ventilation status and eye irritation did not achieve statistical significance ($\chi^2 = 11.27$, $p = 0.080$). The significant association with respiratory problems is consistent with the established pathophysiological role of inadequate ventilation in increasing airborne particulate and chemical concentrations within the breathing zone, thereby amplifying mucosal irritation and bronchoconstriction. The rest of health problems were non-significant findings. The non-significant finding for eye irritation may reflect the multifactorial etiology of ocular problems in this setting, which may include direct mechanical particle contact not amenable to ventilation modification alone.

3.9 Educational Attainment and PPE Compliance

Table 3.9.1 details the cross-tabulation of workers' educational attainment and their frequency of PPE usage, demonstrating a statistically significant overall association ($chi^2 = 16.879$; $p = 0.010$). Furthermore, to assess the specific directional relationship between these ordinal variables, a Spearman's rank correlation was performed (Table 3.9.2)

Table 3.9.1 – Chi Square table of Education level and PPE Usage

Education Level	PPE: Never	PPE: Sometimes	PPE: Always	Total	χ^2	df	p
Illiterate	45	32	18	95			
Primary	37	45	31	113			
Secondary	37	70	37	144	16.879	6	**0.010**
Higher	5	14	5	24			
Total	**124**	**161**	**91**	**376**			

Table 3.9.2: Spearman's Correlation — Educational Level and PPE Usage

Variable Pair	rs	p-value	N
Educational level — PPE usage frequency	0.143	0.005	376

A statistically significant positive correlation was observed between workers' educational attainment and self-reported PPE usage frequency ($rs = 0.143$, $p = 0.005$), indicating that higher levels of formal education are associated with higher PPE usage frequency. However, the strength of this association was weak. This suggests a modest monotonic relationship between education and occupational safety behaviour, implying that while educational attainment may contribute to improved PPE compliance. These findings indicate that educational and health literacy interventions may complement, rather than replace, workplace- and system-level strategies aimed at improving PPE use in this workforce.

3.10 Screening of Independent Variables

Prior to multivariate modeling, we performed independent-samples t-tests for the continuous variables to evaluate their standalone relationship with the dichotomized health outcomes

Independent-samples t-tests were conducted to identify continuous variables demonstrating significant mean differences between workers with low and high problem burden, prior to entry into binary logistic regression models. The complete t-test results for all six health outcomes and all continuous predictors are presented in the Annexure (Table S1). Selected significant findings are summarized in Table 3.10.1 below.

Table 3.10.1: Summary of Significant Independent-Samples T-Test Results

Outcome	**Predictor**	**Mean (Low)**	**Mean (High)**	**t**	**p-value**
Cough	Age (years)	33.58	36.77	−2.592	0.010
	Dust exposure score	5.42	5.06	2.461	0.014
	Standing hours/day	5.96	5.11	3.414	0.001
Skin Problems	Work duration (years)	12.26	9.80	2.017	0.044
	Dust exposure score	5.42	4.82	3.553	< 0.001
MSK Pain	Age (years)	31.50	39.90	−7.369	< 0.001
	Work duration (years)	9.19	15.89	−6.913	< 0.001
	Standing hours/day	5.86	5.33	2.170	0.031
Headache	Age (years)	33.90	36.82	−2.212	0.028

Noise exposure score	6.14	5.47	2.606	0.010

The t-test results reveal several clinically and statistically compelling patterns. For respiratory cough. An unexpected finding was that higher dust exposure scores and more standing hours were paradoxically associated with lower cough burden — a pattern potentially attributable to survivorship bias, whereby workers in the most heavily dust-exposed or physically demanding roles who remain in the workforce may represent a selected, physiologically resilient sub-population.

For musculoskeletal pain, the magnitude of the mean difference in age (31.50 vs. 39.90 years; $t = -7.369$; $p < 0.001$) and work duration (9.19 vs. 15.89 years; $t = -6.913$; $p < 0.001$) between low- and high-burden groups represents the largest and most clinically significant contrasts observed across all t-test analyses in this study. These effect sizes — exceeding 8 years in age and 6.7 years in work tenure — provide robust support for the biological plausibility of cumulative musculoskeletal loading as the dominant driver of MSK problem burden in this population.

For skin problems, workers with higher perceived dust exposure scores showed significantly greater skin problem burden (mean 5.42 vs. 4.82; $p < 0.001$), consistent with established dermatological sensitization mechanisms from chronic wood dust contact. Tinnitus was significantly associated with lower PPE usage scores in problematic workers (0.09 vs. 0.20; $p = 0.012$), underscoring the direct protective function of hearing protection devices. For headache, both older age and higher noise exposure scores were significantly associated with greater problem burden, supporting a cumulative neurological stress model.

3.11 Binary Logistic Regression Analyses

To evaluate the independent variables of the targeted health problems, binary logistic regression models were constructed. This analytical strategy was deliberately selected for its capacity to simultaneously adjust for a comprehensive set of sociodemographic and occupational covariates that might otherwise obscure the true magnitude of the observed associations. By systematically isolating the primary exposures from these potential confounders, the modeling framework generated Adjusted Odds Ratios (AOR) alongside their corresponding 95% Confidence Intervals (CI). Ultimately, this approach yields highly robust and clinically meaningful epidemiological metrics, offering a precise quantification of risk for each evaluated symptom category.

Table 3.11.1: Binary Logistic Regression — Predictors of Respiratory Cough

Predictor	Crude OR (95%Cl)	p-value	AOR (95% CI)	p-Value
Ventilation	2.45 (1.43 – 4.18)	0.001*	2.02 (1.15 – 3.56)	0.015*
Smoking	0.53 (0.33 – 0.86)	0.010*	0.75 (0.44 – 1.28)	0.299
PPE Usage	0.59 (0.44 – 0.79)	<0.001*	0.63 (0.47 – 0.86)	0.003*
Age (years)	1.03 (1.01 – 1.04)	0.011*	1.01 (0.99 – 1.03)	0.402
Dust Exposure Score	0.82 (0.70 – 0.96)	0.015*	0.91 (0.76 – 1.08)	0.274
Standing Hours	0.85 (0.77 – 0.94)	0.001*	0.91 (0.82 – 1.01)	0.09

A comprehensive multivariable binary logistic regression analysis was rigorously conducted to identify the independent predictors of a high respiratory cough burden among the sampled occupational cohort. This finalized model purposefully incorporated key covariates, specifically including workplace ventilation status, smoking history, PPE usage, chronological age, cumulative dust exposure, and standing hours. The overall predictive model demonstrated a highly statistically significant improvement in fit over the unadjusted null model (chi-square = 33.243, df = 6, $p < 0.001$), successfully explaining approximately 11.7% of the total variance in reported cough problem severity (Nagelkerke R-squared = 0.117). Within the fully adjusted analytical framework, inadequate workplace ventilation emerged as a critical, statistically significant independent predictor of high cough burden: workers in poorly ventilated workshops exhibited more than twice the adjusted odds of experiencing a severe respiratory cough burden compared to counterparts in adequately ventilated environments (AOR = 2.02; 95% CI: 1.15–3.56; $p = 0.015$). In stark contrast to preliminary models, the fully adjusted analysis conclusively revealed that specific occupational and behavioral factors fundamentally superseded general lifestyle factors as the primary drivers of adverse respiratory health. Notably, routine Personal

Protective Equipment (PPE) usage emerged as the single strongest independent protective factor; for every incremental one-unit increase in the frequency of PPE utilization, the adjusted odds of reporting a high cough burden significantly decreased by 37% (AOR = 0.63; 95% CI: 0.47–0.86; $p = 0.003$). Additionally, prolonged daily standing hours demonstrated a marginal, non-significant inverse trend in relation to problem severity (AOR = 0.91; 95% CI: 0.82–1.01; $p = 0.090$). Importantly, while variables such as smoking, age, and total cumulative dust exposure exhibited significant bivariate associations with cough severity in preliminary crude analyses, none achieved statistical significance following rigorous multivariable adjustment. Instead, workplace ventilation and PPE compliance accounted for the vast majority of the explained variance ($p > 0.05$). Consequently, the adjusted odds ratio for smoking was rendered entirely non-significant (AOR = 0.75; 95% CI: 0.44–1.28; $p = 0.299$), strongly indicating that the observed crude effect of smoking is heavily confounded by the underlying variance in critical protective workplace behaviors.

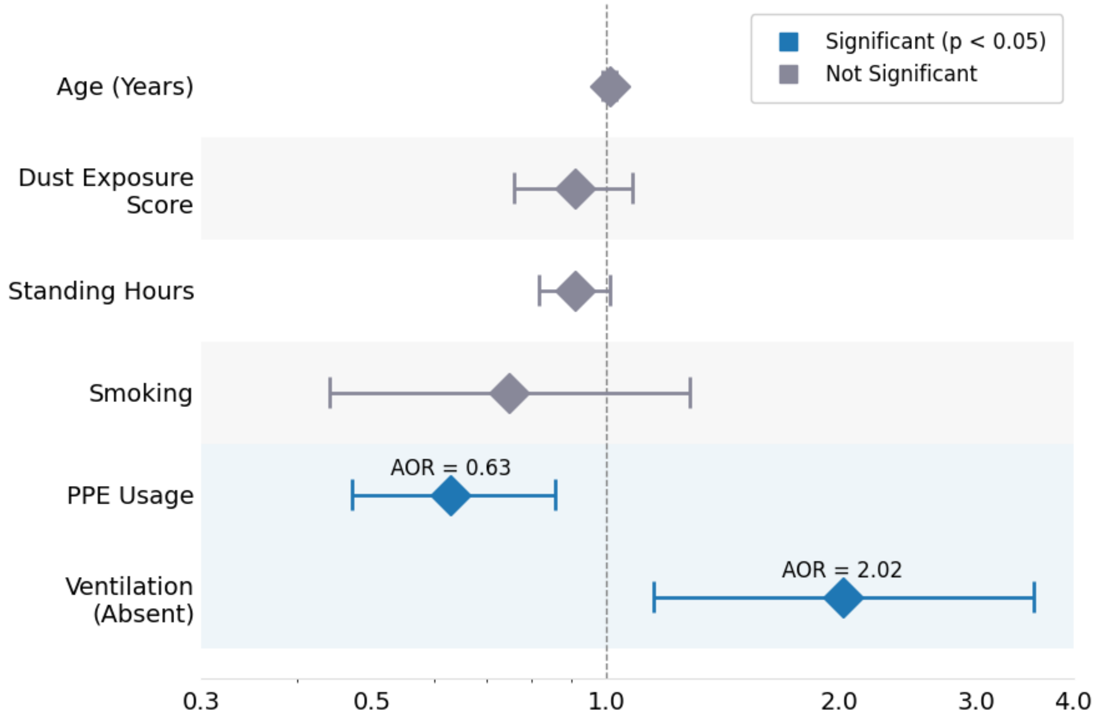

Figure 8: Forest plot of adjusted odds ratios (AOR) for predictors of cough.

3.11.2 Predictors of Musculoskeletal Pain

Table 3.11.2: Binary Logistic Regression — Predictors of Musculoskeletal Pain

Predictor	**Crude OR (95%Cl)**	**p-value**	**AOR (95% CI)**	**p-Value**
Age (years)	1.07 (1.05 – 1.09)	<0.001	1.05 (1.01 – 1.08)	0.005
Total years in wood industry	1.08 (1.05 – 1.10)	<0.001	1.04 (1.00 – 1.07)	0.047
PPE Usage Frequency	0.67 (0.51 – 0.89)	0.006	0.74 (0.54 – 1.01)	0.061
Standing Hours	0.91 (0.83 – 0.99)	0.031	0.98 (0.89 – 1.09)	0.732

A fully adjusted binary logistic regression was conducted to evaluate the predictors of high musculoskeletal (MSK) problem burden. The multivariate model demonstrated exceptional robustness (chi^2 = 56.706, df = 4, p < 0.001), explaining 19.0% of the variance in MSK pain (Nagelkerke R^2 = 0.190), making it the most explanatory model fitted in this study. In contrast to respiratory outcomes, chronic biological factors dominated the MSK risk profile. Age emerged as the primary significant independent predictor: each additional year of life conferred a 5% increase in the odds of reporting severe MSK problems (AOR = 1.05; 95% CI: 1.01–1.08; p = 0.005). Cumulative occupational loading, measured by total years worked in the wood industry, demonstrated significance (AOR = 1.04; 95% CI: 1.00–1.07; p = 0.047), reflecting the strong collinearity between natural aging and career tenure in this population. Notably, daily standing hours (p = 0.732) lost all predictive value after multivariate adjustment. Personal Protective Equipment (PPE) usage exhibited a marginally significant protective trend against MSK pain (AOR = 0.74; p = 0.061), though it did not cross the strict threshold for statistical significance."

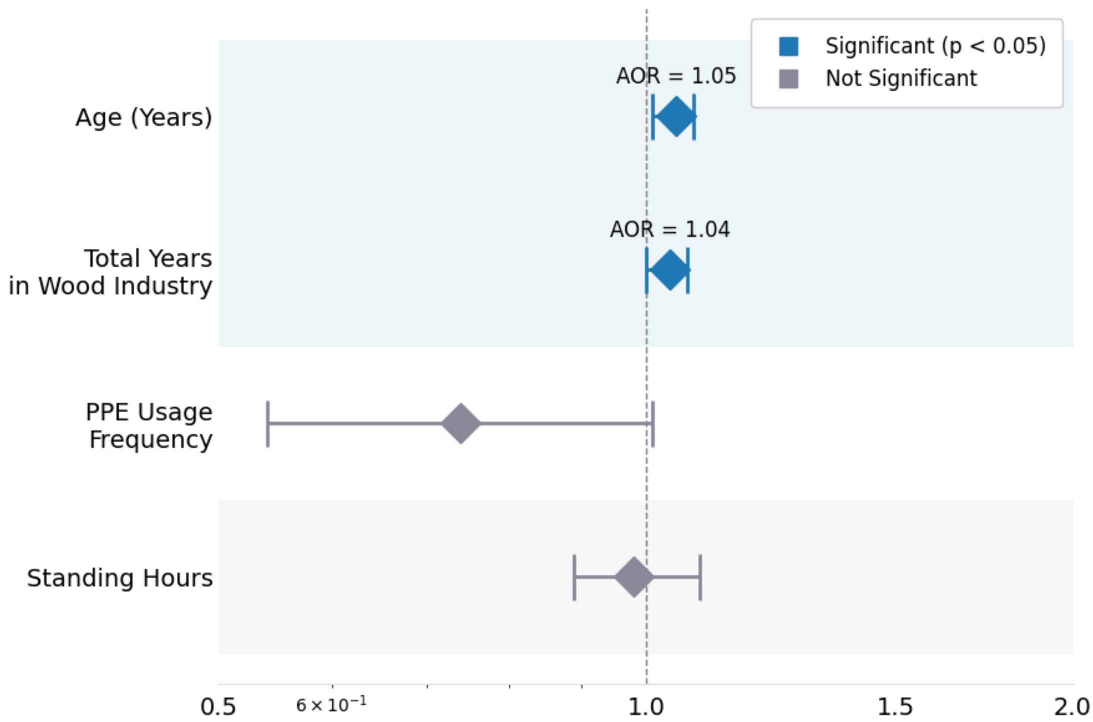

Figure 9: Forest plot of adjusted odds ratios (AOR) for predictors of musculoskeletal pain

3.11.3 Predictors of Skin Problems

Table 3.11.3: Binary Logistic Regression — Predictors of Skin Problems

Predictor	Crude OR (95%Cl)	p-value	AOR (95% CI)	p-Value
Dust Exposure Score	0.92 (0.77 – 1.11)	0.403	0.95 (0.78 – 1.15)	0.577
Total years in wood industry	0.98 (0.96 – 1.01)	0.2	0.98 (0.95 – 1.00)	0.097
PPE Usage	0.60 (0.42 – 0.85)	0.004	0.58 (0.41 – 0.84)	0.003

A fully adjusted binary logistic regression was conducted to evaluate the independent predictors of high skin problem burden. The multivariate model demonstrated a statistically significant improvement over the null model (Chi-square = 17.933, df = 3, $p < 0.001$), explaining 7.3% of the variance in dermatological complaints (Nagelkerke $R^2 = 0.073$). In the adjusted analysis, PPE usage emerged as the sole statistically significant independent predictor of skin problem burden. Each one-unit increase in PPE usage frequency was associated with a 42% reduction in the odds of high skin burden (AOR = 0.58; 95% CI: 0.41–0.84; p = 0.003). Neither dust exposure score (AOR = 0.95; 95% CI: 0.78–1.15; p = 0.577) nor total years in the wood industry (AOR = 0.98; 95% CI: 0.95–1.00; p = 0.097) achieved statistical significance after multivariate adjustment, indicating that the protective role of PPE operates independently of cumulative dust or tenure exposure in determining dermatological outcomes.

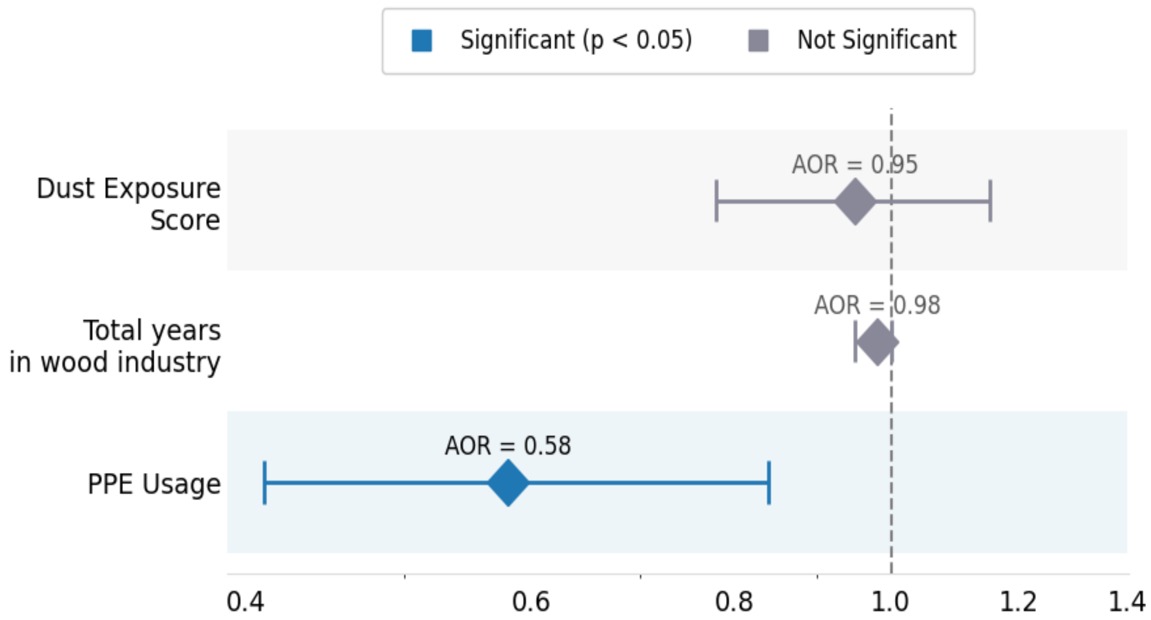

Figure 10: Forest plot of adjusted odds ratios (AOR) for predictors of Skin Problems

3.11.4 Predictors of Headache

Table 3.11.4: Binary Logistic Regression — Predictors of Headache

Predictor	Crude OR (95%Cl)	p-value	AOR (95% CI)	p-Value
Noise Exposure				
Score	0.92 (0.83 – 1.02)	0.122	0.94 (0.85 – 1.04)	0.236
Age (years)	1.02 (1.00 – 1.04)	0.031	1.02 (1.00 – 1.04)	0.065

A binary logistic regression was conducted to evaluate the independent predictors of high headache burden. The adjusted multivariate model demonstrated a statistically significant improvement over the null model (Chi-square = 9.803, df = 2, p = 0.007), explaining approximately 3.7% of the variance in headache frequency (Nagelkerke R^2 = 0.037). In the crude analysis, age demonstrated a statistically significant positive association with headache outcomes (cOR = 1.02; 95% CI: 1.00–1.04; p = 0.031), while noise exposure showed a non-significant inverse trend (cOR = 0.92; 95% CI: 0.83–1.02; p = 0.122). However, after multivariate adjustment, neither predictor achieved conventional statistical significance: age was attenuated to a borderline trend (AOR = 1.02; 95% CI: 1.00–1.04; p = 0.065), and noise exposure remained non-significant (AOR = 0.94; 95% CI: 0.85–1.04; p = 0.236). These findings suggest that while the overall model significantly outperforms the null model, the independent contributions of age and noise exposure to headache burden — when mutually adjusted — do not individually reach the $p < 0.05$ threshold, likely reflecting the modest explanatory power of the model (R^2 = 0.037) and the complex, multi-factorial aetiology of occupational headache in this cohort.

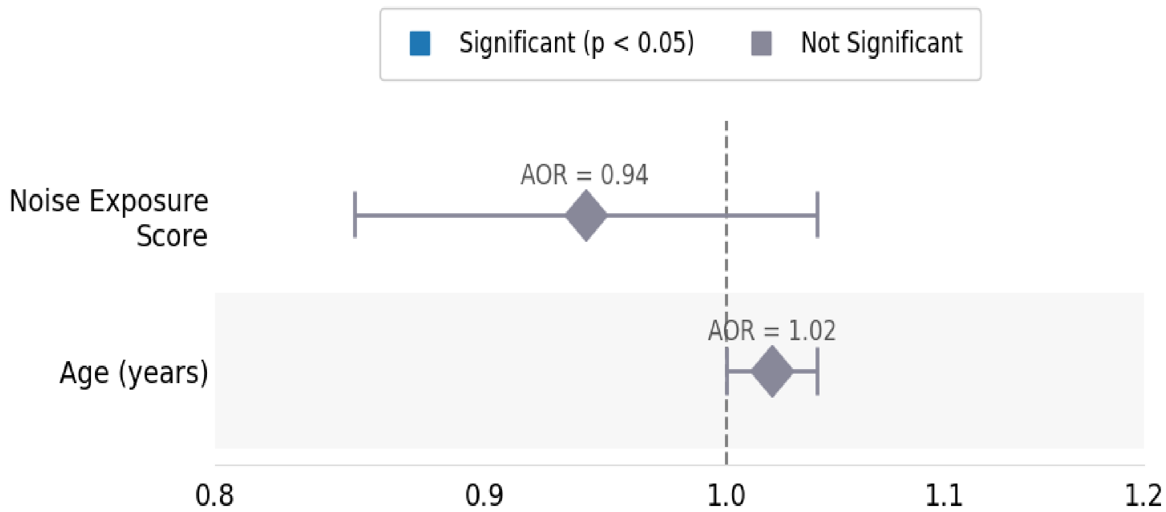

Figure 11: Forest plot of adjusted odds ratios (AOR) for predictors of Headache